

Business assumptions

Average purchase value (\$)
15.00

Cost of sending the better coupon (\$)
2.00

Targeting rule
☒ Value threshold
☐ Budget cap

Filter held-out customers

Coupon type

Bar

Carry out & Take a...

Coffee House

Restaurant(20-50)

Restaurant(<20)

Destination

Home

No Urgent Place

Work

Income bracket

\$100000 or More

\$12500 - \$24999

\$25000 - \$37499

\$37500 - \$49999

\$50000 - \$62499

\$62500 - \$74999

\$75000 - \$87499

\$87500 - \$99999

Less than \$12500

Causal ML for Coupon Targeting

Which customers should receive a 1-day coupon instead of a 2-hour coupon? Every reported causal estimate and targeting result below is computed on a held-out evaluation set, not the rows used to train the models.

Overview & Diagnostics Top Customers Customer Lookup

Showing 3,153 filtered customers from a 3,153-row held-out evaluation set.

Customers to target	Incremental revenue	Campaign cost	Campaign ROI
1,832	\$5,433	\$3,664	48.3%

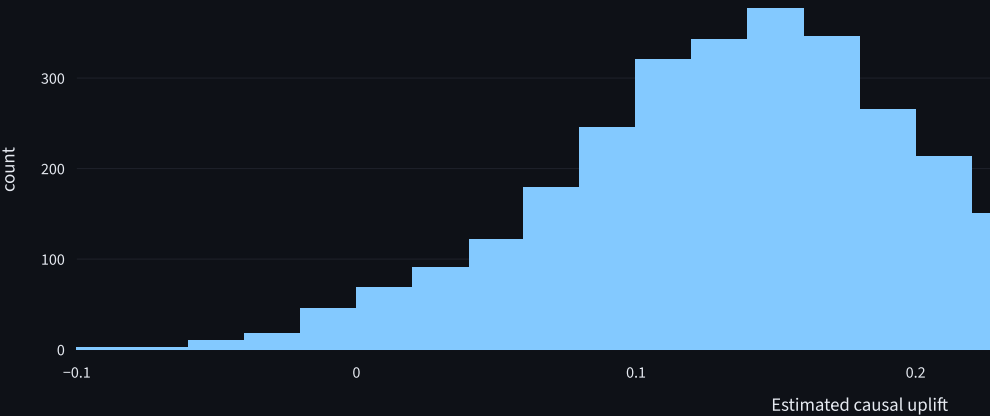
Held-out causal estimates

T-learner average uplift	Propensity matching ATT	Doubly-robust AIPW ATE
+0.148	+0.194	+0.169
↑ 95% CI [+0.129, +0.159]	↑ 95% CI [+0.117, +0.230]	↑ 95% CI [+0.131, +0.199]

The estimators use different assumptions. Agreement is reassuring, but it does not by itself prove that all confounding has been removed.

> Positivity, overlap, and treatment-balance checks

Held-out distribution of estimated individual uplift



Held-out uplift/Qini-style cumulative gain

